Pediatric Nursing Skills



Prepared by/ curricula commetee

Dedication & Acknowledgment

This work is dedicated to the distinguished experts who contributed their sincere efforts and valuable expertise to the development and review of the nursing curricula for Secondary Technical Nursing Schools, playing a pivotal role in refining its academic and practical content to the highest standards. With sincere gratitude and appreciation to:

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WE DEEPLY APPRECIATE THEIR TIME, EFFORT, AND COMMITMENT TO ADVANCING NURSING EDUCATION

Pediatric nursing practice book

N0	Procedures
1	**Neonatal assessment.**
2	**.Neonatal resuscitation**
3	**.O2 therapy**
4	**.Incubator care**
5	**.Infant bath**
6	**Growth measurements**

Physical Assessment of the newborn

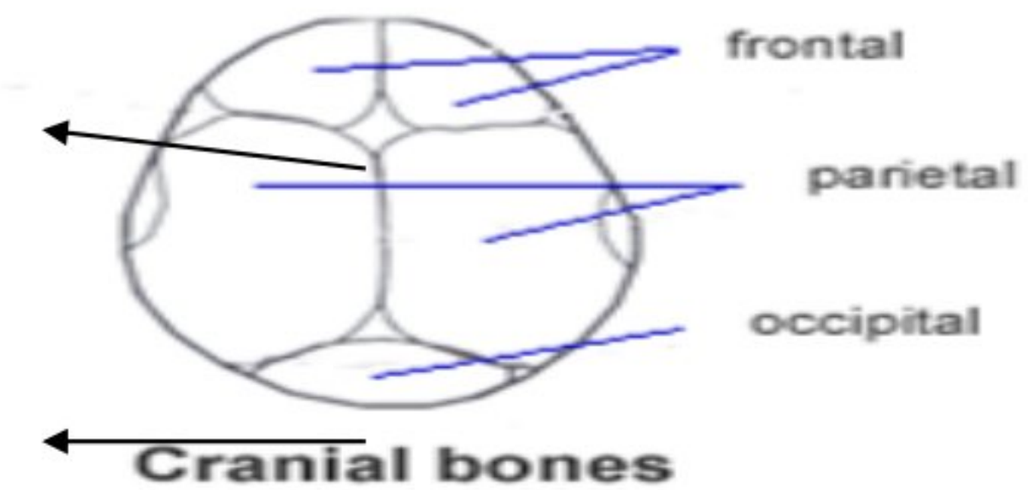
Physical Assessment of the newborn following delivery can be divided into two phases:

- 1- Systemic physical examination.
- 2- The initial assessment using Apgar scoring system.

1-Systematic physical examination:

General appearance:

1-The head:



- The head is $^1/_4$ of the total body length.
- The skull has two fontanel: the anterior and posterior fontanels
- The anterior fontanel is diamond in shape. It is the junction of the sagittal, corneal, and frontal sutures (located between 2 frontal and 2 parietal bones. It is 3-4cm in length and 2-3cm width. It is closes at 12- 18 months of age.
- The posterior fontanel is triangular and is located between occipital and 2 parietal bones. It closes usually by the end of the first month of age.

  Fontanels should be soft, flat, and firm. It bulges when the baby cries or if there is increased intracranial pressure.
- Two conditions may appear in the head. These are caput succedaneum and cephalohematoma.

» ***Caput succedaneum;*** is edema of the scalp resulting from pressure during labor.

» ***Cephalohematoma;*** is a hemorrhage under the periosteum of one of the cranial bones (usually parietal) resulting from trauma of labor.

2-Eyes and ears:

Eyes:

- **Lids**: Usually edematous.
- **Color**: Gray, dark blue, brown. True eyes color is not determined until the age of 3-6 months.
- **Pupil**: React to light, Absence of tears.

Ears:

- **Position**: Top of pinna on horizontal line with outer canthus of eye.
- **Pinna flexible**, cartilage present.

3-Skin:

- At birth, bright red, puffy smooth.
- Second to third day dark pink and dry.
- It is soft and has good elasticity or tissue turgor due to hydrated subcutaneous tissue. Edema is seen around eye, face, legs and scrotum or labia. Cyanosis of hands and feet.
- **Vernix caseosa**: it is a soft yellowish cream, which covers the neonates at birth to protect the skin from infection. It dries within 24-48 hours and rubs off on the infant's clothes.
- **Lanugo hair** it is along soft growths of fine observed on the shoulders, back, extremities forehead of the neonates.
- **Desquamation**: pealing of the skin occurs within 2-4 weeks of life. These are denoted areas where the delicate skin has been rubbed off the nose, knees and elbows, because of pressure and erosion of

sheets. The skin of buttocks is particularly sensitive and should not be left wet and /or soiled.

- **Milia**: these are small pinpoint white or yellow spots due to increased fat secretion. Common on the nose, forehead, cheeks, and chin of the newborn infants.. They will disappear within a few weeks (one to two weeks). They should not be expressed.
    - **Acrocyanosis**: Bluish discoloration on hands and feet first 2-6 hours.
- **Mottling:** Lacy pattern of dilated blood vessels under skin may last several hours to several weeks

4-Gastro-Intestinal Tract and Liver:

- Immature at birth, reaches maturity at years.
- sucking becomes coordinated at 32 weeks
- Little saliva until 3 months of age.

1-The mouth

- Should be examined for abnormalities such as cleft lip and palate

2-The Gum:

- May appear with a quite irregular edge. Sometimes the back of gums may be whitish.

3-Stomach and intestine:

The capacity of an infant's stomach varies after birth from 30-60 cc & increase rapidly. Many infants swallow air during feeding causes

- difficulty in continuing the feed
- vomiting later on
- Colic.

- bowel sounds after 1 hour of birth
- Newborn have difficulty digesting complex starches and fat.
- Abdomen becomes easily distended after eating

- Inspect the umbilical cord for presence of 2 arteries and a vein. Abnormal components may be a pointer to the presence of intra-abdominal anomalies e.g. renal.
- Look for umbilical abnormalities, e.g. hernia,
- Gently palpate the abdomen
- the liver may be palpable up to 2cm below the costal margin
- Stores needed iron for the first few months. Preterm & small infants have lower iron stores than full term and heavier infants (Full term infant's stores last 4-6 month).
- Newborn at risk for **Physiologic Jaundice** after 24 hours of age, related to increased breakdown of RBCs and immature liver functioning.

5-Heart

Apex- fourth to fifth intercostal space, lateral to left sternal border.

6- Respiratory system:-

- Slight sternal retraction is evident during inspiration
- Respiration is chiefly abdominal
- Cough reflex is absent at birth, present in1-2 days.
- First nose breath

What initiates respiration

- Respiration is chiefly Production of lung fluid diminishes 2-4 days before delivery80-100 ml remain in the passageway of a full term infant
- During the birth, fetal chest is compressed and squeezes fluid.
- ***Apnea:*** no breathing for periods of greater than 15 seconds should be evaluated.

***Possible signs of respiratory distress to be aware*:**

- Tachypnea.
- Cyanosis
- Flaring of nostrils
- Grunt
- Retractions

7-Genito-urinary system:-

- Normally , the neonates has urine in his bladder and voids at birth or some hours later
- Neonates pass urine within 24 hours after birth.

initial urine:

- cloudy, scant amounts, uric acid crystals reddish stain on diaper
- Kidneys not fully functional until child is 2 years of age.

***Female genitalia:-**

- Labia & clitoris are usually edematous
- Urethral meatus behind clitoris
- Vernix caseosa between labia

***Male genitalia:-**

- Urethral opening at tip of glans peins
- Tests are palpable in each scrotum.
- Scrotum is usually large edematous, pendulous, covered with rugae & pigmented.

8-The Nervous System:-

- All neurons are present, but many are immature:

Reflex	Eliciting stimulus	Response	Developmental duration
Babinski	Gentle stroke a long sole of foot from heel to toe	Toes fan out, big toe flexes	Disappears by the end of 1st year
Blink	Flash of light or puff of air.	Both eyelids close.	Permanent
Diving reflex.	Sudden splash of cold water in the face.	HR decelerates, blood shunted to brain & heart.	Becomes progressively weak with age.
Knee reflex	Tap on patellar tendon.	Knee kicks	Permanent
Moro reflex OR Startle	Sudden loss of support. Or loud noise	Arms extended, then brought toward each other, lower extremities are extended.	Disappears in about 6 months.
Palmer or grasping reflex	Rod or finger pressed against infant's palm	The object is grasped.	Disappears in 3-4 months.
Rooting reflex	Object lightly brushing infant's cheek	Infant turns towards object & attempts to suck	Disappear 3-4 months

Walking reflex	Infants are held upright & soles of feet are placed on hard surface; baby is dipped slightly forward.	Infant steps forward as if walking.	Disappears3-4 months.
Suckling reflex	Finger or nipple inserted 2 inches into mouth.	Rhythmic suckling	Rhythmic suckling

2- The initial assessment using Apgar scoring system

- It is based on observation of heart rate, respiratory effort, muscles tone, reflex irritability, and color.
- Apgar score is done within the 1st minutes after neonate's body has been born, then is repeated when he is 5 minutes old. Apgar score is assessed by using 5 signs that indicate the state of cardiopulmonary function.
- The total score is 10. if the Apgar score is 4 or below at 1 minute i.e. there is cyanosis, heart rate is below 100 b/m and inadequate or difficult respiration, he will need oxygen. If he has a score of 5-10 he usually needs no treatment.

Score Items	2	1	0
Color	Completely pink body and face	Pink body, blue arms and legs, pale body and face	Pale or blue body and face
Heart rate	>100 heartbeats/min. strong heartbeat.	<100 heartbeats/min. or less weak beats.	No response
Grimace	Crying, coughing and sneezing.	Grimace or puckeima of face.	No response.
Movement and tone	Active movement moving arms and legs	Some movements in response to stimulation	Limp arms and legs. No movements in response to stimulation
Cry and breathing	Strong Regular breathing	Weak cry Slow irregular breathing retraction of chest wall, grunting.	No cry No breathing

Neonatal resuscitation

Learning Objectives:

1- Mention the initial steps of newborn care

2- Demonstrate the insertion of laryngeal mask

3- Perform the insertion of orogastric tube

4- Demonstrate endotracheal intubation

5- perform chest compressions

6- mention medication used during NRP

7- explain resuscitation a stabilization of Babies Born Preterm

8- determine post- resuscitation care

Initial steps of newborn care

1- Time should the umbilical cord be clamped

- ✓ In preterm newborns, potential benefits of delayed cord clamping compared with immediate cord clamping include decreasing the chance of needing medications to support blood pressure after birth, requiring fewer blood transfusions during hospitalization, and possibly improved survival.
- ✓ In term and late preterm newborns, delayed cord clamping may improve early hematologic measurements and, although uncertain, there may be benefits for neurodevelopmental outcomes. However, there may also be an increased chance of needing phototherapy for hyperbilirubinemia.

2- Evaluation the newborn immediately after birth

After birth, all newborns should have a rapid evaluation to determine if they can remain with their mother to continue transition or if they should be moved to a radiant warmer for further assessment. This initial evaluation may occur during the interval between birth and umbilical cord clamping.

You will rapidly ask **3 questions:**

(1) Does the baby appear to be term?

(2) Does the baby have good muscle tone?

(3) Is the baby breathing or crying?





3- The initial steps for *vigorous, term* newborns

If the answers to all 3 rapid evaluation questions are "Yes," the baby can remain with the mother and have the initial steps performed on the mother's chest or abdomen.

- Warmth is maintained by direct skin-to-skin contact
- Dry the baby with the towel or blanket
  - Position the baby on the mother's chest or abdomen
  - If necessary, secretions in the upper airway can be cleared by wiping the baby's mouth and nose with a cloth.
  - After the initial steps are completed, continue monitoring the newborn's breathing, tone, activity, color, and temperature to determine if additional interventions are required.

4- The initial steps for *non-vigorous and preterm* newborns

If the answer to any of the initial evaluation questions is "No," bring the baby to a radiant warmer because additional interventions may be required.

1. **Provide warmth**

Place the baby under a radiant warmer. Leave the baby uncovered to allow full visualization and to permit the radiant heat to reach the baby.

2. **Dry**

Wet skin increases evaporative heat loss. Place the baby on a warm towel or blanket and gently dry any fluid. If the first towel or blanket

becomes wet, discard it and use fresh, warm towels or blankets for continued drying.





3. **Stimulate**

Drying the baby will frequently provide enough stimulation to initiate breathing. If the newborn does not have adequate respirations, brief additional tactile stimulation may stimulate breathing.

If needed, clear secretions from the airway.

- ✓ Brief, gentle suction
- ✓ Suction the mouth before the nose.
- ✓ Be careful not to suction vigorously or deeply.



- ✓ If using a suction catheter, the suction control should be set so that the negative pressure reads approximately 80 to 100 mm Hg when the tubing is occluded.

5- Evaluation the newborn's response to the initial steps

Assess the newborn's respirations to determine if the baby is responding to the initial steps. This should take no more than an additional 30 seconds.

6- Indications of pulse oximetry

- When resuscitation is anticipated.
- To confirm your perception of persistent central cyanosis.
- When supplemental oxygen is administered.
- When PPV is required.

7- Target oxygen saturation

1 min	60%-65%
2 min	65%-70%
3 min	70%-75%
4 min	75%-80%
5 min	80%-85%
10 min	85%-95%

8- The presence of meconium-stained amniotic fluid change

- The presence of meconium-stained amniotic fluid may indicate fetal distress and increases the risk that the baby will require resuscitation after birth.

Insertion of laryngeal mask

- It is intended for use in babies weighing 2 to 5 kg.
- If not already done, attach cardiac monitor leads for accurate assessment of the baby's heart rate.

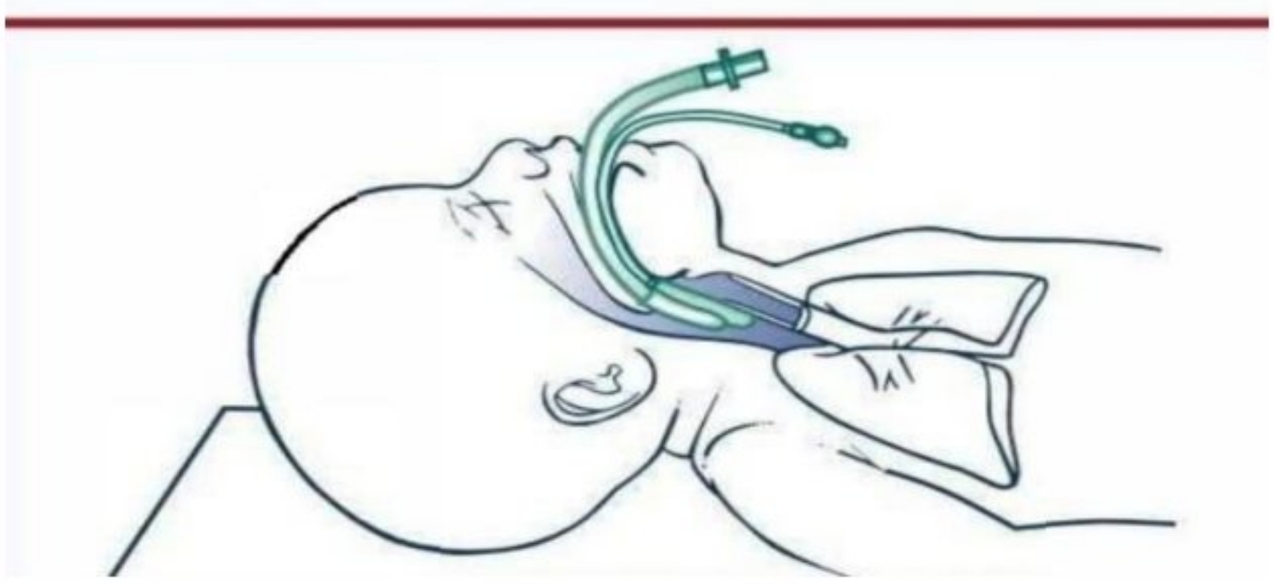
Technique

- Stand at the baby's head and position the baby in the sniffing position.
- Hold the device along the airway tube with the closed bottom of the mask facing the baby's palate and the open bowl of the mask facing toward the baby's chin.
- Open the baby's mouth by pressing gently downward on the baby's chin.
- Insert the leading tip of the mask into the baby's mouth, on top of the tongue, with the bottom of the mask pressed against the baby's palate.
- Glide the device downward and backward, following the contour of the palate, with a continuous but gentle push until you feel definitive resistance.

- Holding the tube in place, attach a Co_2 detector and PPV device. Begin PPV and secure the device in place.
- If the laryngeal mask is correctly inserted and you are providing ventilation that inflates the lungs, you should detect exhaled Co_2 within 8 to 10 positive-pressure breaths. You should see chest wall movement and hear equal breath sounds when you listen with a stethoscope.



Insert laryngeal mask and start PPV



Laryngeal mask forming seal over the glottis

Time of removing laryngeal mask

- ➔ The airway can be removed when the baby establishes effective spontaneous respirations and the device is no longer needed or when an endotracheal tube can be inserted successfully.
- ➔ When you decide to remove the laryngeal mask, suction secretions from the mouth and throat before you remove the device.
- ➔ If the device has an inflatable rim, deflate the rim before removal.

Insertion of orogastric tube

Technique

- ➔ Places the distal end of the orogastric tube at the bridge of the nose and measures to the earlobe and from the earlobe to a point halfway between the xiphoid process and the umbilicus. Notes the centimeter mark on the tube.
- ➔ Inserts the tube through the mouth.
- ➔ Resumes CPAP.
- ➔ Attaches a syringe and removes the gastric contents.
- ➔ Removes the syringe from the tube and leaves the end open as an air vent.
- ➔ Tapes the tube to the baby's cheek.



Measuring the correct insertion depth for an depth tube







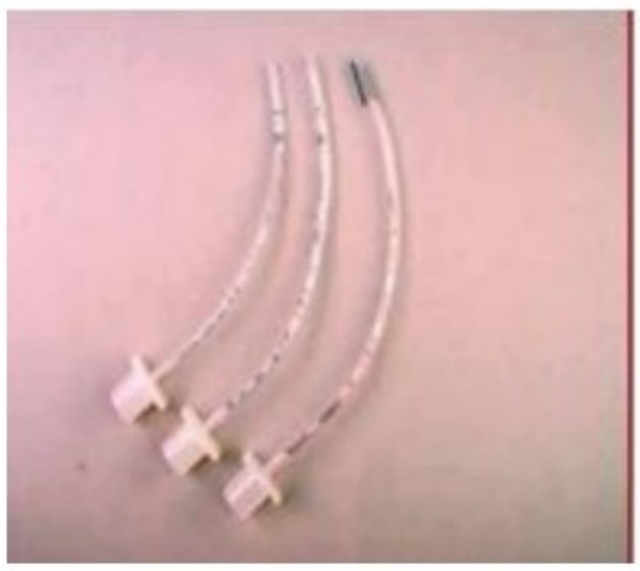


(A) Insertion of an orogastric tube, (B) Aspirating the tube, (C) opening tube to vent, (D) securing tube with tape

Endotracheal intubation

Time of insertion of an endotracheal tube

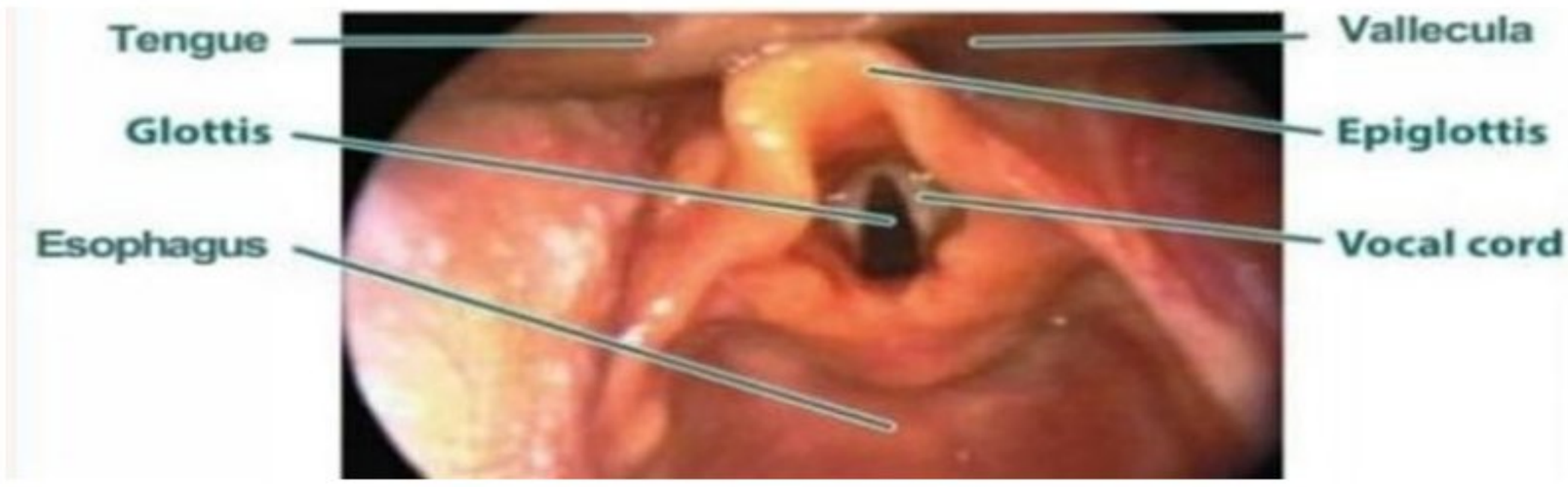
- ☆ recommended if the baby's heart rate remains less than 100 bpm and is not increasing after PPV with a face mask or laryngeal mask.
- ☆ recommended before starting chest compressions. If intubation is not successful or feasible, and the baby weighs more than approximately 2 kg, a laryngeal mask may be used.
- ☆ If PPV is prolonged, an endotracheal tube may be considered to improve the efficacy and ease of assisted ventilation.





Endotracheal tube sizes

Laryngoscope



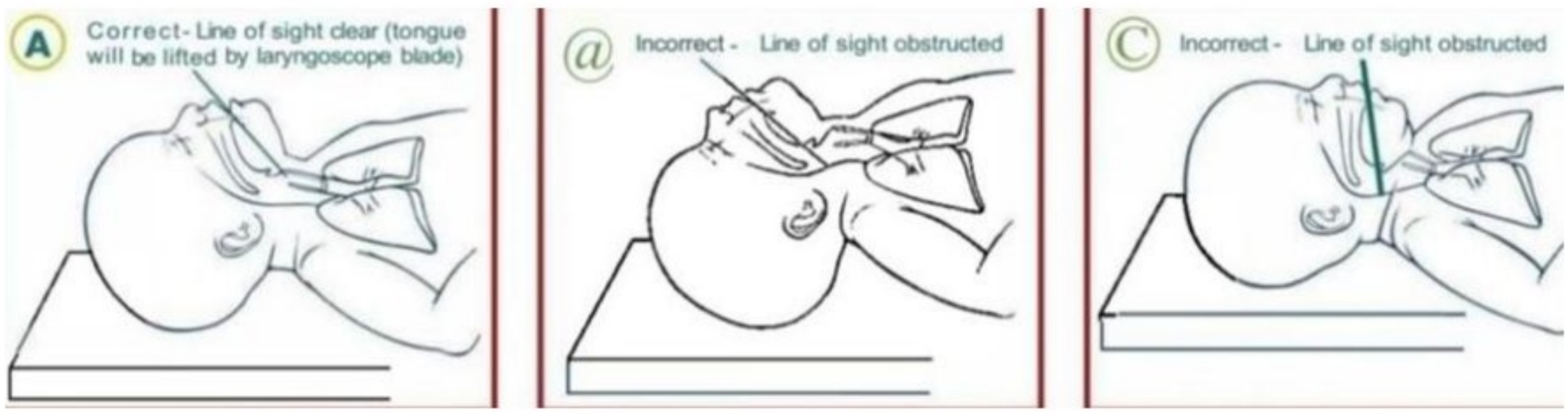
Laryngoscopic view of vocal cords and surrounding structure

Position of newborn for intubation

- Place the baby's head in the midline, the neck slightly extended in the sniffing position, and the body straight. It may be helpful to

place a small roll under the baby's shoulders to maintain slight neck extension.

- This position aligns the trachea for optimal viewing by allowing a straight line of sight into the glottis once the laryngoscope has been properly inserted.
- Adjust the height of the bed, if possible, so that the baby's head is level with the operator's upper abdomen or lower chest to bring the head closer to the operator's eye level and improve the view of the airway.



(A) Correct position, (B and C) incorrect position for intubation

Time allowed for intubation attempt

- The steps of intubation should be completed within approximately **30 seconds**.
- If the baby's vital signs worsen during the procedure (severe bradycardia or decreased oxygen saturation), it is usually preferable to stop, resume PPV with a mask, and then try again.
- Repeated attempts at intubation are not advised because you will increase the likelihood of soft-tissue trauma and make subsequent airway management more difficult.

Confirmation that the endotracheal tube is in the trachea

- The primary methods of confirming endotracheal tube insertion within the trachea are detecting exhaled Co_2 and a rapidly rising heart rate.

Other indicators that the tube is in the trachea

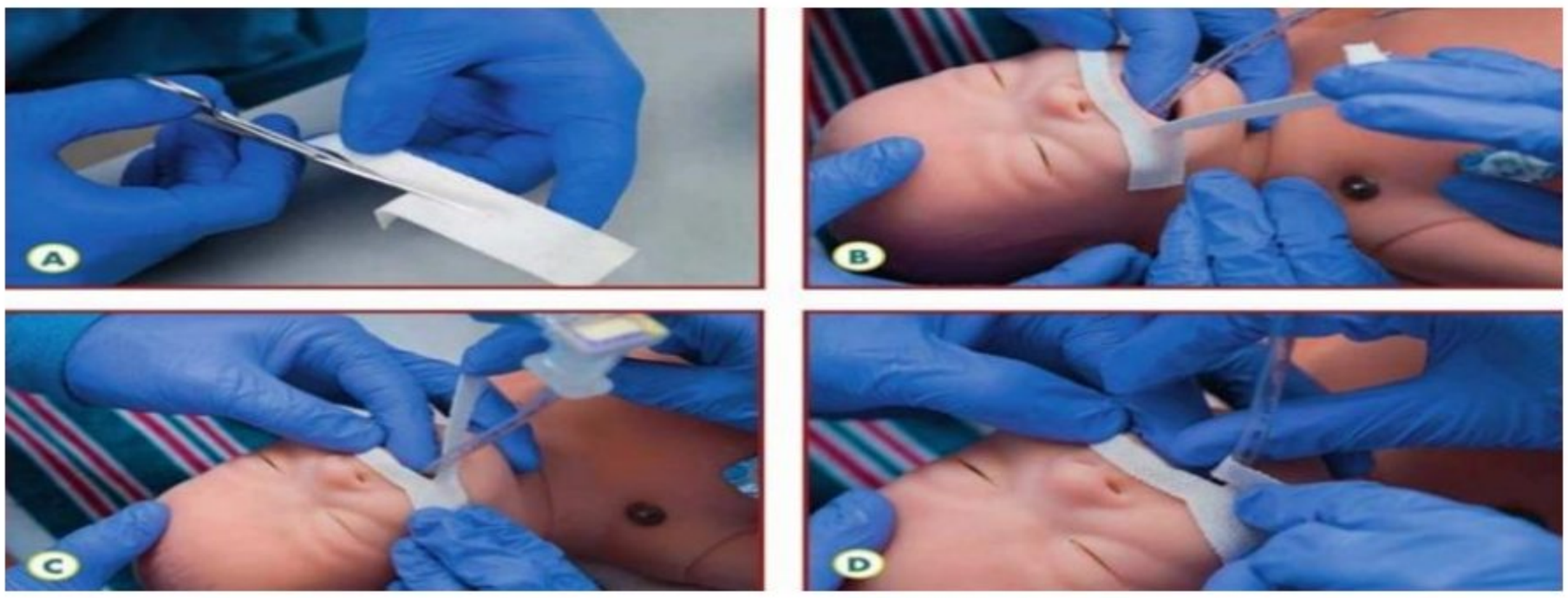
- Audible and equal breath sounds near both axillae during PPV.
- Symmetrical chest movement with each breath.
- Little or no air leak from the mouth during PPV.
- Decreased or absent air entry over the stomach.

If you suspect that the tube is not in the trachea…..

- The tube is not likely to be in the trachea if the Co_2 detector does not show the presence of exhaled Co_2 within 8 to 10 breaths. In most cases, you should remove the tube, resume ventilation with a face mask, ensure that your equipment is properly prepared, ensure that the baby is optimally positioned, and repeat the procedure using a clean tube.
- If the tube position is confirmed and the baby's heart rate does not improve with ventilation through the endotracheal tube, chest compressions are indicated. Once cardiac output is improved, Co_2 will be detected.

Secure tube in place

- After you have correctly positioned the tube, note the centimeter marking on the side of the tube adjacent to the baby's upper lip.
- Cut a piece of 3/4- or 1/2-inch tape so that it is long enough to extend from one side of the baby's mouth, across the upper lip, and about 2 cm onto the opposite cheek.
- Split the tape along half its length so that it appears like a pair of pants.
- Place the un-cut section of tape on the baby's cheek so that the beginning of the split is close to the corner of the baby's mouth. Place the upper leg of tape across the baby's upper lip.
- Carefully wrap the lower leg around the tube.
- At the end, turn the tape onto itself to leave a small "tab" that you can hold to unwind the tape when you want to adjust the insertion depth or remove the tube.
- Listen with a stethoscope over both sides of the chest to be sure the tube has not been displaced. Assess color change on the Co_2 detector and rise and fall of the chest with each assisted breath.
- The tip of the tube should appear in the mid-trachea adjacent to the first or second thoracic vertebra.

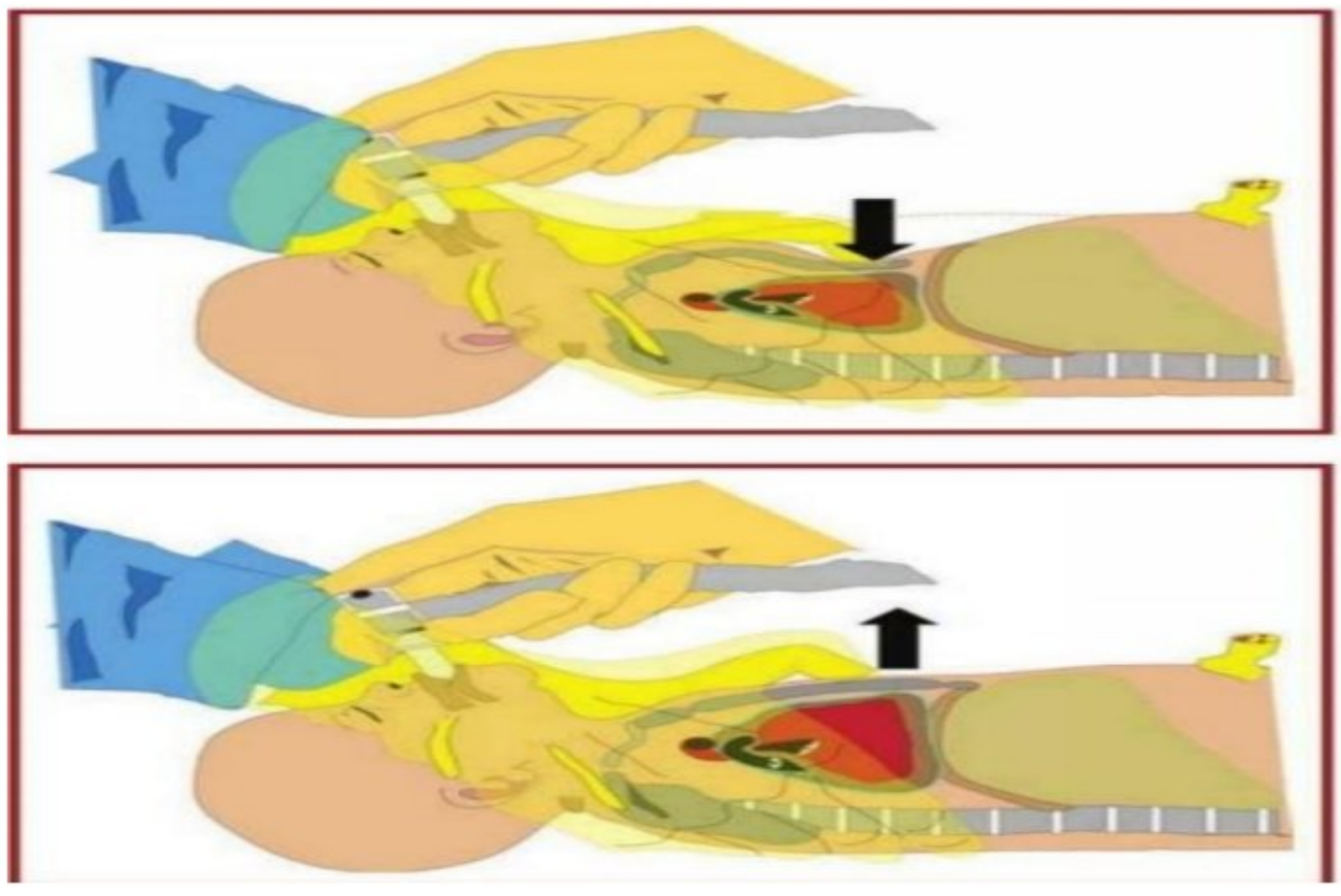


Securing endotracheal tube after insertion

Sudden deterioration after intubation

The DOPE mnemonic	
D	Displaced endotracheal tube
O	Obstructed endotracheal tube
P	Pneumothorax
E	Equipment failure

Chest compressions



Compression (top) and release (bottom) phases of chest compression

Time to begin chest compressions

- Chest compressions are indicated if the baby's heart rate remains less than 60 bpm after at least 30 seconds of PPV that inflates the lungs, as evidenced by chest movement with ventilation.

Right position when administer chest compressions

- When chest compressions are started, you may be standing at the side of the warmer. One of your team members, standing at the head of the bed, will be providing coordinated ventilations through an endotracheal tube.
- Once intubation is completed and the tube is secure, the compressor should move to the head of the bed while the person operating the PPV device moves to the side in addition to providing space for umbilical venous catheter insertion.



Compressor standing at the head of the bed

Position your hands during chest compressions.

- During chest compressions, apply pressure to the lower third of the sternum. Place your thumbs on the center of the sternum, either side-by-side or one on top of the other, just below an imaginary line connecting the baby's nipples. Do not place your thumbs on the ribs or on the xiphoid.



Landmarks for chest compression

Chest compression depth

- With your thumbs correctly positioned, use enough pressure to depress the sternum approximately one-third of the anterior-posterior (AP) diameter of the chest.

Chest compression rate

- The compression rate is 90 compressions per minute. To achieve this rate, you will give 3 rapid compressions and 1 ventilation during each 2-second cycle.

Time to check the baby's heart rate after starting compressions

- Wait 60 seconds after starting coordinated chest compressions and ventilation before briefly pausing compressions to reassess the heart rate.

Time to assess the baby's heart rate response during compressions

- Briefly pause compressions to assess the baby's heart rate. If necessary, briefly pause PPV.
- A cardiac monitor is the preferred method for assessing heart rate during chest compressions. You may assess the baby's heart rate by listening with a stethoscope or using a pulse oximeter.

Time to stop chest compressions

- Stop chest compressions when the heart rate is 60 bpm or greater.
- Once compressions are stopped, return to giving PPV at the faster rate of 40 to 60 breaths per minute.

Interventions if the heart rate is not improving after 60 seconds of compressions

- While continuing to administer chest compressions and coordinated ventilation, your team needs to quickly assess the quality of your ventilation and compressions.
- In most circumstances, endotracheal intubation or laryngeal mask insertion should have been performed.

Medication

Epinephrine

- Epinephrine is a cardiac and vascular stimulant. It causes constriction of blood vessels outside of the heart, which increases blood flow into the coronary arteries.
- Epinephrine increases the rate and strength of cardiac contractions.

Dose of Epinephrine

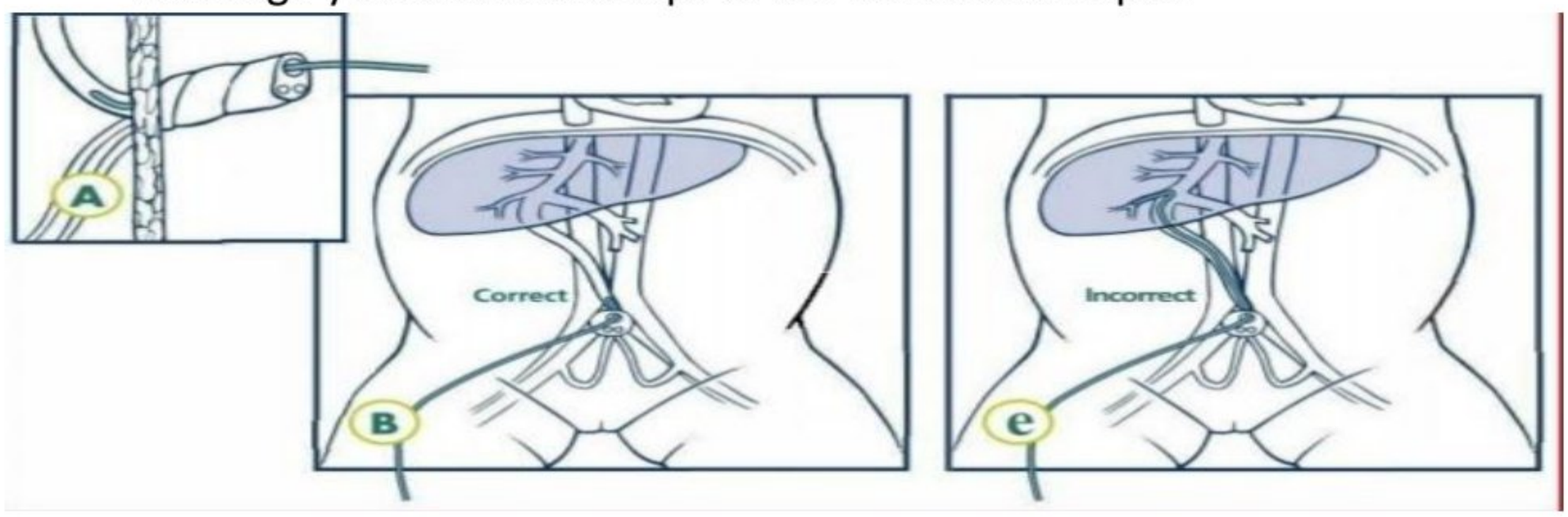
- The only concentration that should be used for neonatal resuscitation is labeled either 0.1 mg/mL or 1 mg/10 mL.

Intravascular access during resuscitation

1. **The umbilical vein**

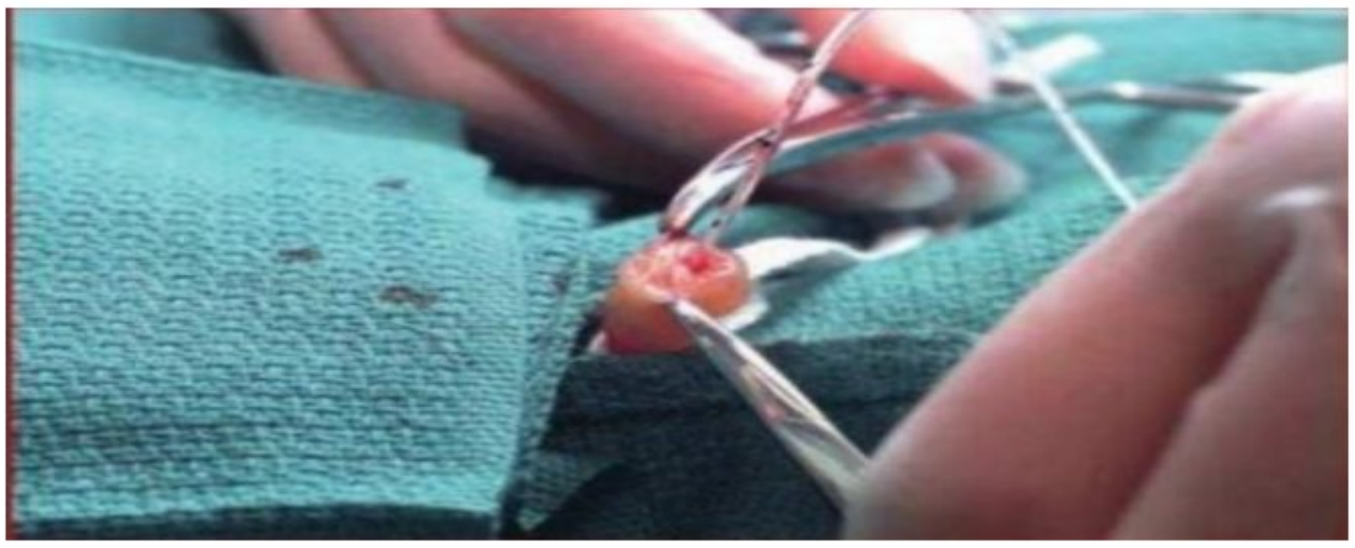
Emergency umbilical venous catheter insertion

- Put on gloves and quickly prepare an area for your equipment Although you should attempt to use sterile technique.



Correct (A , B) and incorrect (C) umbilical venous catheter insertion

- Fill a single lumen umbilical catheter with normal saline using a syringe (3-10 mL) connected to a stopcock. Once filled, close the stopcock to the catheter to prevent fluid loss and air entry



Saline filled catheter inserted into the umbilical vein

- Quickly clean the umbilical cord with an antiseptic solution. Place a loose tie at the base of the umbilical cord around Wharton's jelly or the skin margin.
- Briefly stop chest compressions and caution the team that a scalpel is entering the field. Cut the cord with a scalpel below the umbilical clamp and about 1 to 2 cm above the skin line Attempt to cut straight across the cord rather than at an angle.



Cut the umbilical cord 1 to 2 cm above the skin line

- The umbilical vein will be seen as a larger, thin-walled structure, often near the 12-o'clock position.
- **Insert the catheter into the umbilical vein**

  a. Continue inserting the catheter 2 to 4 cm

  b. for emergency use, the tip of the catheter should be located only a short distance into the vein-only to the point at which blood can be aspirated.

  c. Continue to hold the catheter securely in place with 1 hand until it is either secured or removed.

- Attach the syringe containing either epinephrine or volume expander to the available stopcock port, turn the stopcock so that it is open between the syringe and the catheter.



Open the stopcock toward the baby and infuse the medication

- After medications have been administered, either remove the catheter or secure it for temporary intravenous access as the baby is transported to the nursery.
- If the umbilical venous catheter is not removed, the insertion site should remain uncovered and visible to monitor for bleeding.
- If you remove the catheter, do it slowly and be prepared to control bleeding by tightening the cord tie, squeezing the umbilical stump, or applying pressure above the umbilicus.

2. **The intraosseous needle**
    - Although an umbilical venous catheter is typically the preferred method of obtaining emergency vascular access in the delivery room, an intraosseous needle is a reasonable alternative if umbilical venous access is unsuccessful or not feasible.

Intraosseous needle insertion

- Identify the insertion site. For term newborns, the preferred site is the flat surface of the lower leg, approximately 2 cm below and 1 to 2 cm medial to the tibial tuberosity



Needle insertion site

- Clean the insertion site with antiseptic solution
- Hold the intraosseous needle perpendicular to the skin and advance the needle through the skin to the surface of the bone.
- Direct the needle perpendicular to the bone and advance the needle through the bone cortex into the marrow space.



Insertion using intraosseous drill

- Connect an infusion set (prefilled with normal saline) to the needle's hub, open the stopcock toward the needle, flush the needle with 3 to 5 mL of norn1al saline to open the bone marrow space
- Open the stopcock toward the needle, flush the needle, and infuse the medication or fluid.



Infusion of medication via intraosseous needle

- Monitor the insertion site for evidence of swelling or fluid extravasation.

Resuscitation a stabilization of Babies Born Preterm

Preterm Birth

- Babies born at lower gestational ages are more likely to require additional interventions. Because preterm newborns are also more vulnerable to injury from resuscitation procedures, it is important to find the correct balance between initiating resuscitation without delay and avoiding unnecessarily invasive procedures.

Importance of keeping the preterm newborn warm

- Set the temperature in the room where the baby will be resuscitated and receive initial care to approximately 23°C to 25° C (74° F to 77 ºF).
- Preheat the radiant warmer well before the time of birth.
- After delivery, quickly place a hat on the baby's head.
- Use a pre-warmed transport incubator if the baby will be moved after initial care is completed.
- Maintain the baby's axillary temperature between 36. 5ºC and 37.5 ºC.
-



Thermal mattress placed under blanket on the resuscitation table

Special precautions should be taken after the initial stabilization period

- Monitor the baby's temperature.
- Monitor blood glucose.
- Monitor the baby for apnea and bradycardia.

Post-resuscitation care

Medical conditions may occur in babies who required resuscitation

1. **Temperature instability**
   - After resuscitation, babies may become too cold (hypothermic) or too warm (hyperthermic). Premature newborns are at high risk of hypothermia and this has been associated with increased mortality.
   - Babies may become hyperthermic if their mother has a fever or chorioamnionitis.
2. **Pneumonia and other respiratory problems**
   - The need for resuscitation may be an early sign that a newborn has pneumonia, a perinatal infection, or an aspiration event.
   - Neonatal pneumonia may present with tachypnea and other signs of respiratory distress such as grunting, nasal flaring.

3- Hypoglycemia

- Hypoglycemia may occur because glucose stores are depleted rapidly during perinatal stress. A transiently high glucose level may occur in sorne stressed newborns before the blood glucose level begins to fall.

- Babies who require resuscitation need to have their blood glucose level checked soon after resuscitation and then at regular intervals until it remains stable and within normal limits.
- Intravenous dextrose may be necessary to maintain normal blood glucose levels until oral feedings are established.

4- Seizures or apnea

- Newborns with hypotension, hypoxemia, and acidosis may develop signs of brain injury.
- This injury is called hypoxic-ischemic encephalopathy (HIE). Initially, the baby may have decreased muscle tone, lethargy, poor respiratory effort, or apnea. Seizures may appear after several hours.

O2 therapy administration

Oxygen can be administered through an ET or tracheostomy tube during mechanical ventilation or via an anesthesia bag and mask. For children breathing on their own, oxygen can be delivered via nasal cannula, an oxygen hood or tent, or mask.

Purposes of oxygen therapy:

To relieve hypoxemia results from a respiratory or cardiac emergency or an increase in metabolic function:

1. In a respiratory emergency: Oxygen administration enables the patient to reduce his ventilatory effort.
2. In a cardiac emergency: Oxygen therapy helps to meet the increased myocardial workload as the heart tries to compensate for hypoxemia.
3. When metabolic demand: is high in case of massive trauma, bum, or high fever, oxygen administration supplies the body with enough oxygen to meet its cellular needs.

Methods of Oxygen delivery system:

1. Nasal Cannula
2. Simple Mask
3. Partial Re-Breather Mask
4. Non-Re Breather Mask (NRBM)
5. Venturi Mask
6. Oxygen Hood
7. Oxygen Tent
8. AMBU Bag
9. Tracheostomy Collar, ETT
10. T-piece

Equipment:

- Oxygen-delivery device as ordered by health care provider
- Oxygen tubing (consider extension tubing)
- Humidifier, if indicated
- Sterile water for humidifier
- Face shield as needed for risk of splash
- Clean gloves, if secretions are present
- Oxygen source
- Oxygen flowmeter
- Appropriate “Oxygen in use” signs
- Pulse oximeter
- Stethoscope

Technique of administration of oxygen therapy:

Steps of procedure	Rational
1. Wash hands.	-prevent cross infection.
2. Prepare equipment &supplies.	-save time and effort.
3. Explain to the child and family about O_2 therapy	- to gain their cooperation.
4. Select appropriate methods of oxygen delivery systems.	
5. Set the child in good position (semi fowler's position).	It is appropriate position that facilite breathing
6. Check the cylinder for any leakage for the safety of the patient.	Ensures delivery of prescribed oxygen therapy in conjunction with the specific cannula/mask.
7. Connect flow meter to oxygen source and check the apparatus.	Ensures patency of delivery device and accuracy of prescribed oxygen flow rate.
8. Make sure the humidifier is filled to the appropriate mark.	Humidity prevents drying of nasal and oral mucous membranes and airway secretions.
9. Set flow rate at prescribed liters/minute.	
10. Feel to determine if oxygen is flowing through of cannula.	Flow rate should not exceed 6 L /ruin.
11. Turn on the oxygen.	
12. Check the oxygen flow.	Ensures patency of delivery device and accuracy of prescribed oxygen flow rate.
13. Stress fire hazards involved with oxygen administration.	Put "oxygen in use", "No Smoking" signs outside the child's door and over the bed.
14. Documentation - Date. - Time of staffing and finishing O_2 therapy. - Type of oxygen administration, system used. - The percentage of oxygen delivered and the flow rate. - The child's vital signs, skin color, respiratory effort and lung sounds.	

- The child's response to the procedure and any teaching done with the child or family.	

Nursing considerations

Explain to the parents (and the child, if he's old enough) why oxygen is being administered and how it will help the child's condition. If an ET tube must be used, prepare the child for its insertion, explaining what he'll feel and reassuring him that he'll get used to the tube quickly. In addition:

• Monitor the effectiveness of oxygen therapy by assessing the child's color, pulse oximetry, and PaO2 using ABG analysis.

• Make sure that the patient is receiving the appropriate concentration of oxygen; also make sure that the oxygen is being humidified before delivery to the patient

Recording and Reporting

■ Record respiratory assessment findings; method of oxygen delivery and flow rate; patient's response; any adverse reactions or side effects; any change in health care provider's orders.

■ Report any unexpected outcome to health care provider or nurse in charge.

Care of Incubator



1-Daily Care of Incubator

Equipment:

(Sponge- disinfectant- distilled water- bed sheet- gloves)

Steps of procedure	Rational
1. Wash hands.	-prevent cross infection.
2. Prepare equipment &supplies.	-save time and effort.
3. Wear gloves.	-prevent cross contamination.
4. Replenish humidity tank up to the appropriate level with distilled water.	
5. Wipe the inside wall with disinfectant according to the hospital policy while changing sheet and having neonate in the scale.	
6. Wipe the outside wall every 8 hours with disinfectant.	

7. Wipe the plastic cover of the mattress with disinfectant.	
8. Change the bed sheet.	
9. Monitor oxygen flow rate and concentration as prescribed.	
10. Check that temperature is between 28- 32.	
11. Check that humidity is between 55- 65.	
12. Replace the incubator every 7 days.(date of replacement should be indicated clearly on the incubator)	
13. Place the solid sheet in covered linen hampers.	
14. Remove gloves.	
15. Wash hands.	
16. Recording.	

2-Terminal Care of Incubator

Equipment:

(Sponge or dressing - soapy water- Dilute hypo chloride solution - gloves)

Steps of procedure	Rational
1. Wash hands.	-prevent cross infection.
2. Prepare equipment &supplies.	-save time and effort.
3. Wear gloves.	-prevent cross contamination.
4. Switch electricity off from incubator and the wall socket.	
5. Invert humidity tank to drain it and dry it	

well.	
6. Remove all detachable parts.	
7. Wash the detachable parts with hot soapy water.	
8. Rise and dry it well.	
9. Wash mattress with soap and water, dry it well.	
10. Wash the floor of the incubator with soap and water, dry it well..	
11. Expose the whole unite to adequate fresh air and sunshine.	
12. Change the air filters according to manufacture instruction.	
13. Wipe inside, outside, floor and mattress of the incubator with a dilute hypochlorite solution.	
14. Rinse off the incubator parts and dry it well.	
15. Detach the humidifier and send it to autoclave.	
16. Remove gloves.	
17. Wash hands.	
18. Recording.	

Infant bath



***Purposes**

1-To clean the skin and comfort the child.

2-To stimulate circulation and metabolism.

3-To observe child's abnormalities in the body .

4-To give chance for play and talk with child.

5-To provide an opportunity to note child's growth and development.

6. To give opportunity for parent and infant social interaction.

***Principles**

1-Cleaning one direction by sponge.

3-Eyes, nose and ears require special attention to prevent injury.

2-Decrease child manipulation as possible.

3-Crease, genitalia, buttocks and cord stump must be carefully cleaned (exposing umbilicus to air promotes rapid healing)

4-Bath from inner to outer, from far to near side.

5-Skin should be dried carefully after each moist sponge.

6-Check color and pulse to be sure that the child is tolerating the procedure well.

10-Bath should not be given immediately after feeding due to the risk of regurgitation and aspiration caused by excessive manipulation.

***Equipment**

- Wash basin l for warm water.

- Wash cloth or cotton (sponges) . -lotion.

-Soft hair brush. -Bath towel. - infant clothes. -Plastic bag.

-Mackintosh. -Nail scissors. -Clear water. - Cotton with alcohol.

Steps	**Rational**
1.Wash hands.	To prevent cross infection.
2. Prepare equipment &supplies.	To save time and effort.
3. Explain the procedure to the child's parents.	To Gain cooperation and alleviate fear.
3-Close the doors and windows	To avoid exposure to air current and avoid any respiratory problems or air draft for the child.
4-Assemble the equipment at the child bedside.	To save time and effort and ensure the baby safety.
5-Fill the bowl two-thirds full of water 36.5-40c according to the age and season, check water temperature by thermometer or by elbow joint. (Change the water as needed).	This temperature relaxes muscles, increase circulation and comfortable to the child. -to maintain proper temperature.
6-Bath the child from top to bottom as following	
-Wipe each eye with the moist sponge from the inner to outer aspect. Beginning with the far eye, then dry gently each one.	
- Clean the infant `s face and wipe around her/his	

mouth and nose (and inner if discharge), then go over her/his cheeks and forehead, dry with sponge.	
- Clean each ear, don`t try to wipe inside : just go over and behind it.	
-Clean the head (scalp) and dry it.	
-Wipe the neck, thoroughly with particular attention given to creases.	
-Take-off child`s clothes.	
- Clean hands, from fingers then clean the hands and dry well.	
-Wash trunk and dry it.	
-Clean the chest and abdomen then dry well in one direction.	
Turn infant on near side wash back, dry area with towel and cover trunk after drying.	
-Clean the lower extremities from bottom to top then dry well.	
-Clean diaper area (according to diaper care procedure).	
7.Place the infant into the basin slowly , insert feet first.	To not frighten him.
8.Remove the infant from basin and dry the skin well especially skin folds	
9. Dress the infant ..	
10.Clean the finger nails and toes nails	
clean the equipment	
Recording :- time , observation & infant reaction	

Growth measurements

Introduction

Measurement of physical growth in children is a key element for evaluation the children health status.

purpose:

1. To assess the health status of children.
2. To detect any abnormalities early in children.
3. To assess nutritional state of children.
4. To determine the dosage of medication "Weight'.

Growth measurements include:

1. Height/Length.
2. Weight
3. Head circumference.
4. Chest circumference.
5. Upper arm circumference.
6. Skin fold thickness .

1.Length recumbent length

Length was measured when children are in supine position, or recumbent position and are less than 2 years old.(the average new born length at birth is 50cm).

Equipment:

Measuring board, measuring tape, hard surface.





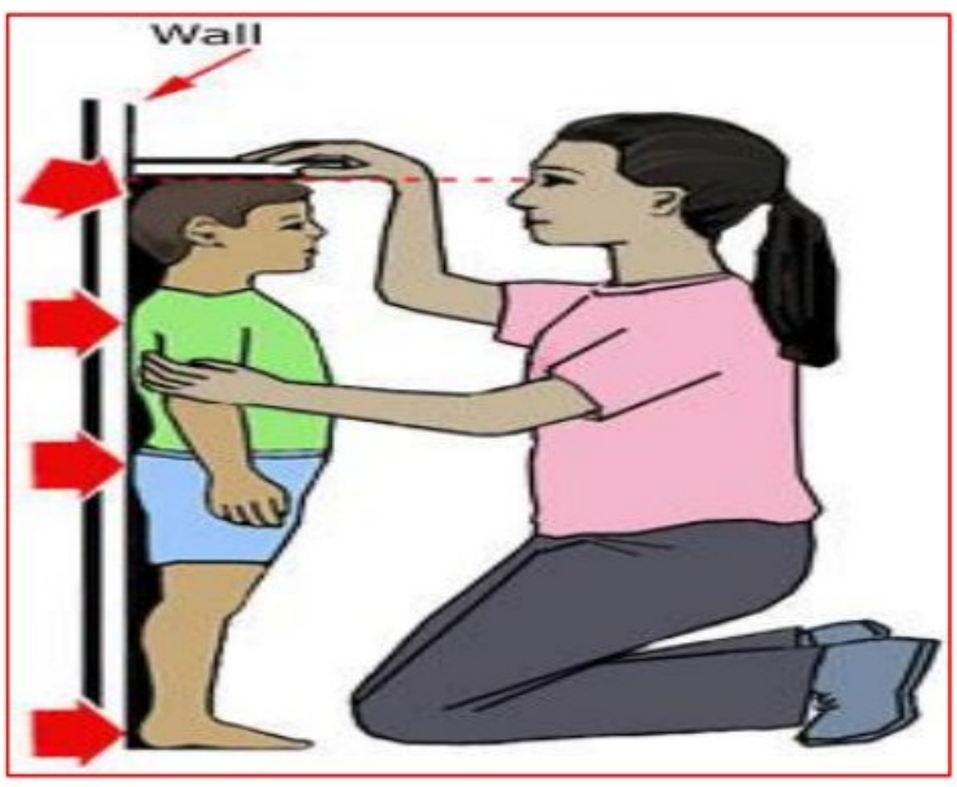
Procedure:

Steps	Rational
1. Wash hands.	prevent cross infection.
2. Prepare equipment &supplies.	save time and effort.
3. Explain the procedure to the child's parents	Gain cooperation and alleviate fear.
4.Keep privacy & prepare environment	
5. place the towel on the board.	prevent cross contamination.
6. Remove the infant's socks and cap.	For accurate measurement.
7. Place the infant on the center of the board in supine position.	Normal position during infancy is flexed.
8. place the head against the head board firmly and food against food board .	

9. Grasp the knees together gently.	
10. Push down on the knees until the legs are fully extended and hold the legs firmly.	For accurate measurement.
11. Bring the headboard against the soles of heels firmly.	
12. Read and record.	
13. Return and clean the reused equipment.	
14. Wash hands.	
15. Recording.	
If a measuring device is not available length is taken.	
1. place the infant on a paper covered surface.	
2. Push down the knees.	-legs become fully extended.
3. Make points of the top of the head and heal of the feet with a pen.	
4. Remove the child from his place.	
5. Measure between these two points.	
6.Recording.	

2- Height

Height was measured when children are standing position and are more than 2 years of age





Equipment:

Measuring board for old child – stadiometer.

Procedure

Steps of procedure	**rational**
1. Wash hands.	prevent cross infection.
2. Prepare equipment &supplies.	save time and effort.
3. Explain the procedure to the child and/or child's parents.	Gain cooperation and alleviate fear.
4.Keep privacy & prepare environment.	
1. Remove the infant socks and shoes .	prevent any error in height.
2. Stand as tall and straight as possible with head in the midline and the line of vision parallel to the floor.	For accurate measurement.
3. The child back should be to vertical flat surface with heels, buttocks and back of the shoulder at one line and eye looking straight as possible .	Normal position during infancy is flexed.

4. Any flexion of the knees, slumping of the shoulder or raising of the heels of the feet is checked and corrected.	
5.Move the board on the top of the head.	
6. Read and record to the nearest centimeter.	
7. Return and clean the reused equipments.	
8. Wash hands.	
9. Recording.	
If a measuring device is not available , height is taken by using measuring tape	
1. Attach a measuring tape to the wall.	-prevent any error in height.
2. The child adjusted to the tape.	-For accurate measurement.
3. Place a three-dimensional object, such as a thick book or box, on top of the head.	
4.The side of the book must rest firmly against the wall to form a right angle.	
5.Move the board on the top of the head.	
6. Read and record to the nearest centimeter.	

3- Weight

The average new born weight at birth is 3kgm .

A- Infant's weight

Equipment:

1. Appropriate sized beam balance scale.
2. Scale paper.
3. Cotton with alcohol.
4. Pen and paper.



Steps	Rational
1. Wash hands.	-prevent cross infection.
2. Prepare equipment &supplies.	-save time and effort.
3. Explain the procedure to the child and/or child's parents.	-Gain cooperation and alleviate fear.
4.Keep privacy & prepare environment.	
5. Place the scale horizontally on a firm surface	
6.Balance the scale (setting it at zero point).	-To prevent error on weight.
7. Wipe the scale using cotton with alcohol.	-prevent contamination.
8. Put baby on the scale.	
9. Warm the environment.	-For accurate measurement.
10. Remove the infant clothing	- prevent cross contamination
11. place the infant in the center of scale .	
12.Support the infant without touching.	-to measure accurately
13. Adjust the reading and record	-To plot the reading on percentile chart.

b- children weight (over 2 years old)





Equipment:

- Standing scale.

– Paper , pen .

Steps of procedure	**Rational**
1. Wash hands.	-prevent cross infection.
2. Prepare equipment &supplies.	-save time and effort.
3. Explain the procedure to the child and/or child's parents.	-Gain cooperation and alleviate fear.
4.Keep privacy & prepare environment.	
5. Place the scale horizontally on a firm surface	
6. check to see that setting it at zero point.	-To prevent error on weight.
7. Put scale paper on the scale for the child to stand on.	- prevent cross contamination
8.Warm the environment	
9. Remove the child's clothing .	-For accurate measurement.
10. tell the child to stand and keep erect on the scale	
11. read the scale measurement.	
12. Return and clean the reused equipments.	
13. Wash hands.	
14. Recording.	

4. Head circumference

(New born head circumference at birth is between 33- 35cm).

Head circumference should be measured:

-Under 36 months of age.

-It is important to detect any neurological defect.



Equipment:

1. Measuring tape.
2. Cotton with alcohol.
3. Paper and pen.

Steps of procedure	**Rational**
1. Wash hands.	- prevent cross infection.
2. Prepare equipment &supplies.	- save time and effort.
3. Explain the procedure to the parents.	- Gain cooperation and alleviate fear.
Clean the measuring tape with cotton and alcohol.	To prevent the infection.
4. Pass the tape from the frontal area bone above the eye brows and pinna of the ears, and around the occipital prominence at the back of the skull	- This is the greatest head circumference
5. read the result.	
6. Return and clean the reused equipments.	
7. Wash hands.	
8. Recording.	

5. Chest circumference

The normal new born chest circumference at birth is 30.5 to 33cm.

Equipment:

1. Measuring tape.
2. Cotton with alcohol.
3. Paper and pen

Steps	Rational
1. Wash hands.	-prevent cross infection.
2. Prepare equipment &supplies.	-save time and effort.
3. Explain the procedure to the child and/or child's parents.	-Gain cooperation and alleviate fear.
4. Remove the child's clothing of upper half.	- To fix the measuring tape well.
5. Place on the flat table in a supine position for infant, stand alone for children.	
6. Place the tape across the nipple line.	
7. Take measurement between inspiration and expiration.	
8. Record the average of the two values.	
9. Return and clean the reused equipments.	
10. Wash hands.	
11. Recording.	

Note:

- Head circumference > chest circumference……At birth.
- Head circumference = chest circumference……12 – 24 month.
- Head circumference = chest circumference……24 month.

6) Upper arm circumference.

Arm circumference is an indirect measure of total muscle mass.

Equipment :

Measuring tape.

Steps of procedure	Rational
1. Wash hands.	-prevent cross infection.
2. Prepare equipment &supplies.	-save time and effort.
3. Explain the procedure to the child or child's parents.	-Gain cooperation and alleviate fear.
4. With the child 's right arm flexed 90 degree at elbow , mark midpoint between a cromion and olecranon process on posterior aspect of arm .	- To measure accurately
5,Wrap the measuring tape around upper arm at midpoint.	
9. Return and clean the reused equipments.	
10. Wash hands.	
11. Recording.	

7.Skin fold thickness

Skin fold thickness is an index of total body fat . skin fold thikness can be taken at: abdomen ,or upper thigh or upper arm.

Equipment :

Special caliper.

Steps of procedure	Rational
1. Wash hands.	-prevent cross infection.
2. Prepare equipment &supplies.	-save time and effort.
3. Explain the procedure to the child or child's parents.	-Gain cooperation and alleviate fear.
4.Mark midpoint as done before.	- To measure accurately
5,with arm hanging freely ,grasp a fold of skin between thumb and finger 1cm above midpoint.	
6.Gently pull fold away from underlying muscle and continue to hold until measurement is completed.	
7.Place caliper over skin fold at mid point mark.	
8. Estimate reading to nearest 1mm ,2to 3 seconds after applying pressure.	
9. Return and clean the reused equipments.	
10. Wash hands.	
11. Recording.	